

variables.

- Minor** : The 'minor' of any element in a determinant is another determinant of lower order that can be obtained by deleting the row and column containing that element. So, in a 3rd order determinant minor of any element will be a 2nd order determinant.
- Rank** : Rank is the maximum number of independent rows (or, columns) in a matrix.
- Trace** : The trace of a square matrix is the sum of its diagonal elements.

5.6 ANSWERS/ HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) -9

2) Do yourself.

$$\begin{aligned}
 3) \quad & \text{Hint : } \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} \\
 & = \begin{vmatrix} 1 & 0 & 0 \\ a & b-a & c-a \\ a^2 & b^2-a^2 & c^2-a^2 \end{vmatrix} \begin{cases} c_2^1 = c_2 - c_1 \\ c_3^1 = c_3 - c_1 \end{cases} \\
 & = 1 \begin{vmatrix} b-a & c-a \\ b^2-a^2 & c^2-a^2 \end{vmatrix} \text{ [Expanding by rows]} \\
 & = (a-b)(b-c)(c-a)]
 \end{aligned}$$

$$4) \quad \left[\text{Hint: given determinant} = \begin{vmatrix} (b+c)-(b+c) & a-(2c+a) & a-(2b+a) \\ b & c+a & b \\ c & c & a+b \end{vmatrix} \right]$$

$$(R_1' = R_2 + R_3 - R_1)$$

$$= -2 \begin{vmatrix} 0 & c & b \\ b & c+a & b \\ c & c & a+b \end{vmatrix} = -2 \begin{vmatrix} 0 & c & b \\ b & a & 0 \\ c & 0 & a \end{vmatrix} \begin{cases} R_2'' = R_2 - R_1' \\ R_3'' = R_3 - R_1' \end{cases}$$

$$= +2 \begin{vmatrix} 0 & b & c \\ b & 0 & a \\ c & a & 0 \end{vmatrix} = 4abc \text{ (after evaluating)}$$

Check Your Progress 2

1) Do yourself after reading Section 5.2.

$$2) \left[\text{Hint: } \Delta = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 1 \\ 3 & 2 & 9 \end{vmatrix} = -20 \neq 0 \quad \therefore \Delta_1 = \begin{vmatrix} 6 & 2 & 3 \\ 7 & 4 & 1 \\ 14 & 2 & 9 \end{vmatrix} = -20 \right.$$

$$\Delta_2 = \begin{vmatrix} 1 & 6 & 3 \\ 2 & 7 & 1 \\ 3 & 14 & 9 \end{vmatrix} = -20 \quad \Delta_3 = \begin{vmatrix} 1 & 2 & 6 \\ 2 & 4 & 7 \\ 3 & 2 & 14 \end{vmatrix} = -20$$

$$\therefore x = \frac{\Delta_1}{\Delta} = 1, \quad y = \frac{\Delta_2}{\Delta} = 1, \quad z = \frac{\Delta_3}{\Delta} = 1 \quad]$$

Check Your Progress 3

1) [Hint: $|A| = 1(36-25) - 2(12-5) + 3(5-3) = 11-14+6 = 3 \neq 0$

$$\therefore \text{Ad } A = \begin{bmatrix} 11 & -9 & 1 \\ -7 & 9 & -2 \\ 2 & -3 & 1 \end{bmatrix} \therefore A^{-1} = \frac{\text{Adj. } A}{|A|} = \frac{1}{3} \begin{bmatrix} 11 & -9 & 1 \\ -7 & 9 & -2 \\ 2 & -3 & 1 \end{bmatrix}$$

3) Rank is less than 3

Check Your Progress 4

1) See 5.3.5.1

2) See 5.3.5.3

3) First partition the matrices A and B then, separately determine $[A_{11}B_{11} + A_{12}B_{21}]$, $[A_{11}B_{12} + A_{12}B_{22}]$, $[A_{21}B_{11} + A_{22}B_{21}]$ and $[A_{21}B_{12} + A_{22}B_{22}]$

Check Your Progress 5

1) Follow the example in sub-section 5.3.7

2) Refer to sub-section 5.3.7

3) Refer to sub-section 5.3.7

5.7 EXERCISES

Q1. Use Cramer's rule to solve the following national income model:

$$Y = C + I + G \quad \text{And} \quad C = a + bY$$

[Hint: Find Y and C, consider I, G, a, b, as constants]

$$\text{Ans: } Y = \frac{I+G+a}{1-b}, \quad C = \frac{a+b(I+G)}{1-b}$$

Q2. Consider the following national income determination model:

$$Y = C + I + G, \quad c = a + b(Y - T), \quad T = d + tY$$

Where Y (national income), C (consumption expenditure) and T (tax collection) are endogenous variables, I (investment) and G (government expenditure) are exogenous variables; t is the income tax rate.

Solve for the endogenous variables, using Cramer's rule.

$$\left[\begin{array}{l} \text{Ans : } y = \frac{bd - (I + G + a)}{b(1-t) - 1}, \quad c = \frac{-a + bd - b(1-t)(I + G)}{b(1-t) - 1}, \quad T \\ = \frac{-d + bd - at - (I + G)t}{b(1-t) - 1} \end{array} \right]$$

Q3. The following is an example of the Keynesian model with the money.

Behavioral equations are

$$C = 0.8Y, \quad I = 102 - 0.2r,$$

$$M^d = 0.25Y - 2.5r, \quad M^s = 100$$

Equilibrium conditions are $Y = C + I$ and $M^d = M^s$

- ii) Write down the equations of the IS and LM curves.
- iii) Evaluate the equilibrium value of Y and r using Cramer's rule.

Ans. $Y = 500, r = 10$

Q4. A set of linear equations in which the right-hand side consisting of constants has zeros, is called a set of homogeneous equations.

Show that $ax + by + cz = 0, bx + cy + az = 0, cx + cy + bz = 0$

$$\text{with } D = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

has the trivial solution ($x = y = z = 0$) if $D \neq 0$ and has a non-trivial solution if $a^3 + b^3 + c^3 - 3abc = 0$.

Ans. Find the solution treating x to be constant.

Q5. Consider the national income model given by

$$Y = C + I + G$$

$$c = a + b(Y - T) \quad (a > 0, \quad 0 < b < 1)$$

$$T = d + tY \quad (d > 0, \quad 0 < t < 1)$$

Ans. Solve this model for the variables Y, C and T by matrix inversion.

Q6. If $U = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}$, $V = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $X = \begin{bmatrix} 4 & 2 & -1 \\ -1 & 3 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ and $Y = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

then find a. UV ; b. $VU+X$; c. XY

Ans. Use partitioned matrices to check the results.



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UNIT 6 VECTOR ANALYSIS

Structure

- 6.0 Objectives
- 6.1 Introduction
- 6.2 Vector Addition and Scalar Multiplication
- 6.3 Geometrical and Physical Interpretations
- 6.4 Norm and Inner Product
- 6.5 Vector Spaces and Subspaces
- 6.6 Linear Dependence of Vectors
- 6.7 Generators and Basis
- 6.8 Let Us Sum Up
- 6.9 Key Words
- 6.10 Answers/Hints to Check Your Progress Exercises
- 6.11 Exercises

6.0 OBJECTIVES

After going through this unit, you should be able to

- identify the presentation of variables with magnitude and direction;
- apply the basic operations of addition, subtraction and multiplication of vectors;
- present geometrical and physical interpretation of vectors and vector operations;
- explain the concepts of norm, inner product, vector space, linearly independent and linearly dependent set of vectors; and
- explain the concepts of generator and basis of a vector space.

6.1 INTRODUCTION

Generally, the elements of a set can be arranged in any manner. It does not matter whether a three-element set is written as $\{a, b, c\}$ or as $\{a, c, b\}$ etc. But for certain purposes this is not helpful and we need to introduce the consideration of order. Such sets are distinguished not only by their members but also by the order in which elements appear. Thus, as an '*ordered*' set, the pair (a, b) is different from (b, a) for $b \neq a$. An ordered set of this kind is called a '*vector*'.

Here we shall consider only vectors with a finite number of elements. Further, unless stated otherwise, we shall assume that each element in a vector is a real number. Thus, we are typically concerned with the n -tuples $(x_1, x_2, \dots, x_n)$ of real

numbers. A vector, all of whose elements are zero, is called the '**null vector**'. In economic analysis, quantities of various commodities consumed by an individual can be represented as a vector.

6.2 VECTOR ADDITION AND SCALAR MULTIPLICATION

A given vector, i.e., an ordered set of finite number of elements, can be written as

a row, e.g., $(a, b, c, \dots)$ or, as a column, e.g., $\begin{pmatrix} a \\ b \\ c \\ \vdots \end{pmatrix}$.

These are referred to as **row and column vectors** respectively.

Let us consider a vector $\mathbf{a} = (a_1, a_2, \dots, a_n)$ in which each $a_i \in \mathbf{R}$. Then the vector $\mathbf{a}$ is said to be a vector over the field $\mathbf{R}$. Like ordinary numbers, **sum and difference of two vectors** can be defined provided they have the same number of elements.

Thus, if, $\mathbf{a} = (a_1, a_2, \dots, a_n)$; and $\mathbf{b} = (b_1, b_2, \dots, b_n)$

then $\mathbf{a} + \mathbf{b} = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$ and

$\mathbf{a} - \mathbf{b} = (a_1 - b_1, a_2 - b_2, \dots, a_n - b_n)$

Note that here $\mathbf{a} = \mathbf{b}$ if and only if $a_1 = b_1, a_2 = b_2, \dots, a_n = b_n$.

Further, if λ is a real number, we can define **scalar multiplication** $(\lambda \cdot \mathbf{a})$ of vector $\mathbf{a}$ by a scalar, (Note '**scalar product of two vectors**' is a different concept to be discussed in section 5.4), as follows:

$$(\lambda \cdot \mathbf{a}) = (\lambda a_1, \lambda a_2, \dots, \lambda a_n)$$

Example: $\mathbf{a} = (1, 3, -2)$, $\mathbf{b} = (4, 0, 6)$

Then $\mathbf{a} + \mathbf{b} = (1+4, 3+0, -2+6) = (5, 3, 4)$;

$\mathbf{a} - \mathbf{b} = (1-4, 3-0, -2-6) = (-3, 3, -8)$ and $3\mathbf{a} = (3, 9, -6)$

Check Your Progress 1

- 1) List the difference between vector and scalar presentation.

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- 2) If $\mathbf{a}_1 = (2, 3, 4, 7)$, $\mathbf{a}_2 = (0, 0, 0, 1)$ and $\mathbf{a}_3 = (1, 0, 1, 0)$, find $\mathbf{a}_1 + 2\mathbf{a}_2 + 3\mathbf{a}_3$.

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- 3) If $x_1 = (2, 9, 8)$, $x_2 = (0, 1, 0)$ and $x_3 = (1, 0, 1)$, find $2x_2 + 5x_3 - x_1$.

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6.3 GEOMETRICAL AND PHYSICAL INTERPRETATIONS

Vector is often represented geometrically by a line with an arrowhead on it (say, $\vec{r}$). The length of the line indicates the **magnitude of the vector**, and the arrow denotes its **direction**. It should be noted that a vector is not just a number. If we are considering vector lying in a plane, then two numbers are needed to describe any vector: one for its magnitude and another giving its direction (the angle it makes with one of the coordinate axes). If vectors in three-dimensional spaces are being studied, three numbers are needed to describe any vector: one number for its magnitude, and two to denote its orientation with respect to some coordinate system.

In general, a vector may originate at any point in space and may terminate at any point. Physical quantities such as force are, of course, independent of where one places the vector in a coordinate system. They depend only on the magnitude and direction of the vector. For this reason, it is convenient to have a vector start always at the origin of the coordinate system as in Figure 6.1. *Here we shall, for simplicity adopt the convention that all vectors begin at the origin of the coordinate system rather than at some other point in space.*

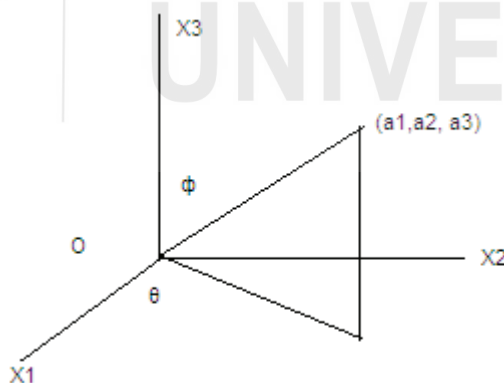
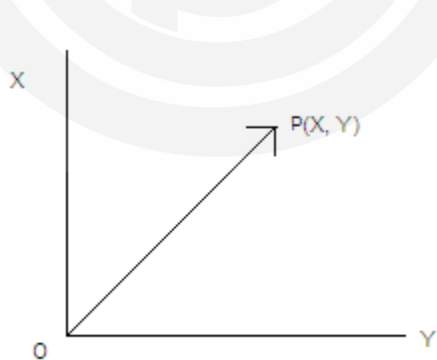


Fig. 6.1: Vector starting from the origin

Fig. 6.2: Characterised Vector

Consider a vector in Figure 6.2. It will be observed that by specifying point (a_1, a_2, a_3) , that is, the point where the head of the vector terminates, we have completely characterized the vector. Its magnitude (in some physical units) is $\sqrt{a_1^2 + a_2^2 + a_3^2}$ and its direction is characterised by the two angles θ and ϕ , where

$$\tan\theta = \frac{a_2}{a_1} \quad \text{and} \quad \cos\phi = \frac{a_3}{\sqrt{a_1^2 + a_2^2 + a_3^2}}.$$

Thus, there is a one-to-one correspondence between all points in space and all vectors, which emanate from the origin. For any given point (a_1, a_2, a_3) , a corresponding unique vector can be drawn from the origin to this point. Conversely, for any given vector $\mathbf{a}$, there is a unique point (a_1, a_2, a_3) , which is the point where the vector terminates. Because of this correspondence between vectors and points in 3-dimensional space, we can write $\mathbf{a} = (a_1, a_2, a_3)$.

6.4 NORM AND INNER PRODUCT

Both through scalar multiplication and vector addition transform one vector, or, a pair of vectors into *another vector*. Unlike these concepts defined in section 6.2, we will now define functions, which transform a vector into a *scalar*, a real number.

For a pair of vectors (the same length, say, n):

$\mathbf{a} = (a_1, a_2, \dots, a_n)$ and $\mathbf{b} = (b_1, b_2, \dots, b_n)$, we define their *scalar product* (as against scalar multiplication of section 6.2 as

$$\langle \mathbf{a}, \mathbf{b} \rangle = a_1 b_1 + a_2 b_2 + \dots + a_n b_n = \sum_{i=1}^n a_i b_i$$

Scalar product is also called *inner product* and even *dot product* of two vectors.

Example: If $\mathbf{a} = (1, -1, 2)$; $\mathbf{b} = (-2, 3, 6)$,

then $\langle \mathbf{a}, \mathbf{b} \rangle = (1) \cdot (-2) + (-1) \cdot (3) + (2) \cdot (6) = 7$

Norm of a vector: As the first step towards finding the norm of a vector, let us consider a given vector $\mathbf{a} = (a_1, a_2, \dots, a_n)$, and find out inner product of $\mathbf{a}$ with itself. The inner product would be $\|\mathbf{a}\|^2 = a_1^2 + a_2^2 + \dots + a_n^2$. Now, look at the following example.

Example: If $\mathbf{a} = (5, -2, 3)$ then $\|\mathbf{a}\|^2 = 5^2 + (-2)^2 + 3^2 = 38$.

The positive square root of the inner product of a vector with itself is called the *norm or length (or magnitude, as stated earlier) of the vector* $\mathbf{X} = (x_1, x_2, \dots, x_n)$ and is denoted by $\ell(\mathbf{X})$.

Thus, $\ell(\mathbf{X}) = \|\mathbf{x}\|^{\frac{1}{2}} = [\sum_{i=1}^n x_i^2]^{\frac{1}{2}}$

In the case of the above example, the norm would be $\sqrt{38}$.

Clearly, the norm defined above can be regarded as the '*inner product*' of vector $\mathbf{a}$ with itself.

Two vectors are said to be *orthogonal* to each other if their inner product is zero, i.e., $\langle \mathbf{a}, \mathbf{b} \rangle = 0$,

e.g., if $\mathbf{a} = (1, 3)$, $\mathbf{b} = (-3, 1)$, then $\langle \mathbf{a}, \mathbf{b} \rangle = 0$.

It is easy to see that that null vector $\mathbf{0}$ is orthogonal to every other vector.

Using the concepts of distance and the inner product, we can now define

$$\cos \theta = \frac{\langle \mathbf{a}, \mathbf{b} \rangle}{\{\|\mathbf{a}\| \cdot \|\mathbf{b}\|\}^{\frac{1}{2}}}$$

By Schwartz's inequality, it is shown that $\left(\sum_{i=1}^n a_i b_i\right)^2 \leq \sum_{i=1}^n a_i^2 \sum_{i=1}^n b_i^2$
 $\Rightarrow \cos^2 \theta \leq 1$ as it should be.

Two results follow right away from the above definition.

a) If $\mathbf{b} = \lambda \mathbf{a}$ i.e., $\mathbf{b}$ is a scalar multiple of $\mathbf{a}$,

then $\|\mathbf{b}\| = \lambda^2 \|\mathbf{a}\|$ and $\langle \mathbf{a}, \mathbf{b} \rangle = \lambda \|\mathbf{a}\|^2 \Rightarrow \cos \theta = 1$ or $\theta = 0$

b) If $\langle \mathbf{a}, \mathbf{b} \rangle = 0$ i.e., $\mathbf{a}$ and $\mathbf{b}$ are orthogonal,

then $\cos \theta = 0 \Rightarrow \theta = \pi/2$. In other words, $\mathbf{a}$ and $\mathbf{b}$ are perpendicular to each other.

Check Your Progress 2

1) Given $x = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$ and $y = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$, find $2y$ and $x-y$ graphically.

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2) Find the norm of the following vectors: (i) (2, 5); (ii) (-2, 2).

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3) Find the inner product of the following pairs of vectors:

i) (2, 3, 4) and (4, 5, 5)

ii) (-2, -3, 4) and (4, 5, -6).

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6.5 VECTOR SPACES AND SUBSPACES

Let us now define a set $\mathcal{V}$ such that it contains, as its elements, all n -component vectors that can be generated from the field of real numbers, i.e.,

$$\mathcal{V} = \left\{ (x_1, x_2, \dots, x_n) \mid x_i \in \mathbf{R}; i = 1, 2, \dots, n \right\}$$

Properties:

i) If $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathcal{V}$, then $\mathbf{x} + \mathbf{y} \in \mathcal{V}$

Further, we have

$$\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}, \text{ and,}$$

$$\mathbf{x} + (\mathbf{y} + \mathbf{z}) = (\mathbf{x} + \mathbf{y}) + \mathbf{z}.$$

$\mathcal{V}$ contains a vector $\mathbf{0}$ such that $\mathbf{x} + \mathbf{0} = \mathbf{x}$

ii) For any $\lambda, \theta, \delta \in \mathbf{R}$, and for $\mathbf{x}$ and $\mathbf{y}$ in $\mathcal{V}$, it must be true that $\lambda \cdot \mathbf{x} \in \mathcal{V}$

such that $\lambda \cdot (\delta \cdot \mathbf{x}) = (\lambda \cdot \delta) \cdot \mathbf{x}$, $\lambda \cdot (\delta \cdot \mathbf{x}) = \delta (\lambda \cdot \mathbf{x})$ and $1 \cdot \mathbf{x} = \mathbf{x}$

iii) $\lambda(\theta + \delta)\mathbf{x} = (\lambda\theta + \lambda\delta)\mathbf{x}$ and $\lambda(\mathbf{x} + \mathbf{y}) = \lambda\mathbf{x} + \lambda\mathbf{y}$

The set $\mathcal{V}$ satisfying i, ii and iii above is said to be a **vector space or a linear space over the field $\mathbf{R}$** .

Summarizing: The aforesaid properties imply that $\mathcal{V}$ is closed under vector addition as well as under scalar multiplication of vectors. Both operations of vector addition and scalar multiplication are commutative and associative. Scalar multiplication is distributive over vector addition. And $\mathcal{V}$ contains an element $\mathbf{0}$, the null vector, in which each component is equal to zero.

Definition: A nonempty subset w , along with the two operations of vector addition and scalar multiplication, is said to be a **subspace of $\mathcal{V}$** if

a) w is a subset of $\mathcal{V}$ and,

b) w is a vector space in its own right, i.e., it satisfies all the axioms specified above for a vector space.

Example: Consider a two-dimensional plane, $\mathcal{V}_2$, and suppose the set w consisting of all points lying along the horizontal (or the vertical) axis. w is a subset of $\mathcal{V}_2$ and it satisfies all the properties listed above. In fact, all points on any line drawn through the origin constitute a subspace. Algebraically, each of the following

$$w = \{(x_1, x_2) \mid x_1 \in \mathbf{R}, x_2 = 0\},$$

$$w = \{(x_1, x_2) \mid x_2 \in \mathbf{R}, x_1 = 0\}$$

$$\text{and } w = \{(x_1, x_2) \mid c_1 x_1 + c_2 x_2 = 0\}$$

$$\text{is a subspace of } \mathcal{V}_2 = \{(x_1, x_2) \mid x_1 \in \mathbf{R}, x_2 \in \mathbf{R}\}$$